JAVIER CRUZ VILLARREAL

Madrid, Spain | Monterrey, Mexico | EU Citizenship | Eligible to sign internship agreement via IE University
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SUMMARY

Computer Science & AI student who builds AI and machine learning products and leads small cross-functional teams from idea to demo. CTO across AI startups and client projects, hackathon winner, and most at home close to the customer, bridging engineering, product, and business. Targeting Forward Deployed Engineer, AI Product Manager, and Technical Program Manager roles.

EXPERIENCE

Hospitia, CTO | AI Hotel Inspection Platform (stealth) *Sep 2026 – Present*

- Own the technology for a **four-stage** AI vision pipeline that automates hotel brand-compliance inspections and PIP reports.
- Leading the move to production on **AWS** (Bedrock, S3) and benchmarking open-weight vision models (Gemma, Qwen) on accuracy and cost, to standardize on a single model family.

IEU Labs x Sybol, Software Engineer & Team Lead | AI Compliance Engine *Mar 2026 – Aug 2026*

- Led a **7-person team** across technical, research, and QA workstreams, setting architecture and milestones with Sybol's CTO.
- Architected an **agentic RAG pipeline** over EU AI Act, GDPR, and Digital Product Passport regulation, returning structured compliance rules per media asset.
- Designed a media authenticity scoring engine (**0–1 scale**, deepfake to authentic) issuing signed **W3C Verifiable Credentials**.

Pharmable, Co-founder & CTO | AI Platform for Pharma Regulatory Affairs *2026*

- Co-founded an AI platform for pharma regulatory affairs; currently building the MVP, leading product scoping and technical architecture for Series A–C biotechs preparing **FDA/EMA** filings, shaped through customer discovery with regulatory affairs leads.

Extern x Outamation, AI Document Intelligence Extern *Dec 2025 – Feb 2026*

- Selected for the IE x Extern program (**12% acceptance**); built OCR + LLM + RAG pipelines turning multipage PDFs into structured JSON.

PROJECTS & COMPETITIONS

SnapFrost, Frost Early-Warning for Andean Potato Farmers | BCG Platinion x Welthungerhilfe **1st Madrid, 2nd EMEA** | *Oct 2026*

- Placed **1st in Madrid** and **2nd in the EMEA final** against the winners of four BCG Platinion hubs, building a Raspberry Pi 5 sensor station with SMS alerts in **2 days**.
- Quantile gradient-boosting model cut forecast error **14%** vs the raw ECMWF forecast on **358 unseen nights** (1.50 to 1.29 °C).
- Prototype keeps working offline (local alarm, buffered readings) and warns farmers by **SMS** on basic phones; proposed a one-season pilot with a municipality and a farmers' association.

Agentic Wildfire Crisis Response | HackSpain 2026 x HappyRobot **1st Place, HappyRobot Track** | *Sep 2026*

- Won the HappyRobot track (1 of **5 tracks**, **~250 builders**) in **36 hours** with a **4-person team**; AI agents dispatch drones and trucks on a custom fire simulator with incomplete information.
- Built parts of the bilingual operator control-room interface, including the dual-map crisis view.
- Simulation engine validates every agent order, and a hidden second ignition stays unseen until a vehicle spots it, so agents have to act on partial information.

StreamScope, Real-Time Twitch Chat Intelligence Platform **2nd / 25 teams** | *IE Tech Venture Bootcamp*

- Placed **2nd of 25 teams**, mostly Master's students; led product and a **4-person team** from idea to working demo and pitch, building Flask REST APIs, an NLP classification pipeline, and a live analytics dashboard.

IATA x Infosys Sustainability Datathon **Top 5 / 38 teams** | *Oct – Dec 2025*

- Cost-benefit model on IATA, EUROCONTROL, and Eurostat data; proposed a levy cutting cost per ton of CO2 from **$633 to $29 (95%)**.

LEADERSHIP & ACTIVITIES

- **IE Technology & Innovation Club**, Partnerships Officer: helped organize **TechIE Day** and its Cursor hackathon; lining up **HappyRobot** as partner for the next edition and managing outreach to tech companies and startups.
- **AI Lab Ambassador**, IE University (selected for performance).
- **Content:** short-form videos with **5M+ organic views** on Instagram and **20M+** on TikTok; managed UGC creators and local TikTok/IG accounts for the Bump (amo) Mexico launch.

EDUCATION

IE School of Sciences and Technology, B.S. Computer Science & Artificial Intelligence (3rd year) *Madrid | Expected Jun 2028*

Tecnológico de Monterrey, B.S. Computer Science and Technology *Monterrey | Sep 2022 – Jan 2024*

SKILLS

AI & ML: LLMs, RAG, Agentic AI, Prompt Engineering, Machine Learning, Computer Vision, NLP, OCR, Vector Search, Forecasting, MCP
Engineering: Python, SQL, TypeScript, React, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, AWS, GCP, Docker, Git, CI/CD
Product & Delivery: Product Management, Project Management, Stakeholder Management, Client-Facing Delivery, Team Leadership
Languages: Spanish (Native), English (C1)